

Voice-Based Access Control in Virtual Reality Using Natural Conversational Speaker Verification

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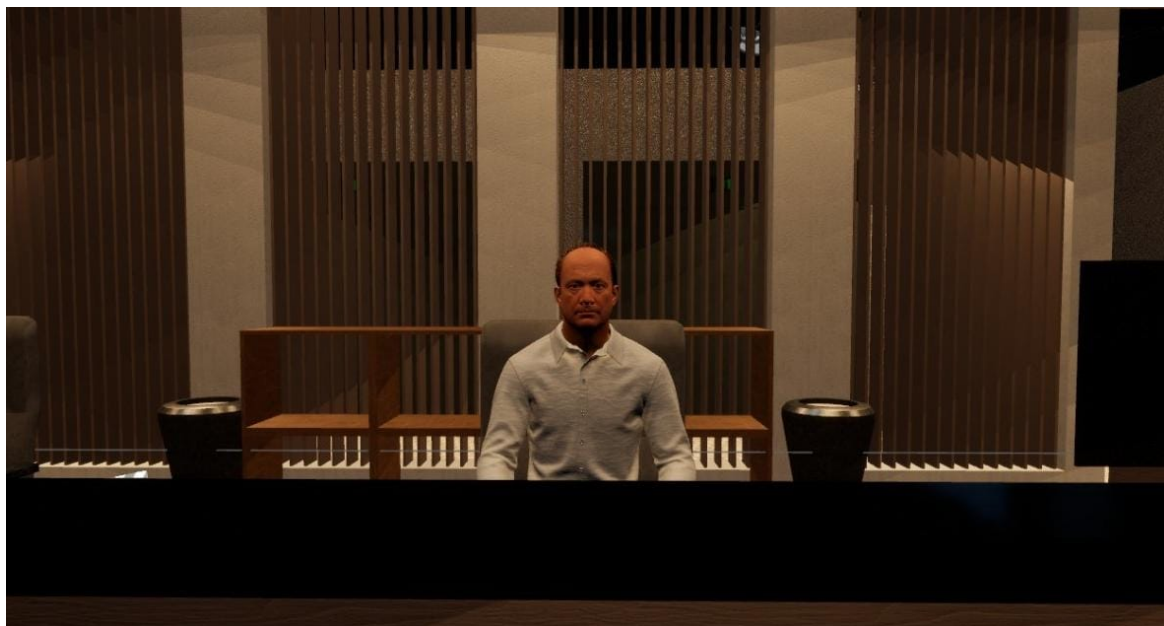


Figure 1: Demo scenario showing the virtual entrance guarded by a non-player character (NPC).

ABSTRACT

Voice-based biometric authentication offers a natural and hands-free alternative for access control in immersive Virtual Reality (VR) environments. We present a VR-based access control demo in which a non-player character (NPC) acts as a virtual gatekeeper, authenticating users through conversational speaker verification. Instead of explicit enrollment procedures or predefined passphrases, the system leverages a short, natural greeting exchange to perform speaker verification and, when necessary, seamless enrollment. The demo illustrates how voice biometrics can be integrated into VR in a user-friendly manner, preserving immersion while enabling secure and personalized access control.

Index Terms: Virtual Reality, Speaker Verification, Conversational Interaction, Access Control

1 INTRODUCTION

Authentication remains a fundamental challenge in immersive environments, where traditional mechanisms such as passwords, PINs, keyboards, or explicit biometric scanners often disrupt user immersion and reduce usability. In Virtual Reality (VR), interaction

paradigms benefit from modalities that are natural, continuous, and hands-free, motivating the exploration of voice as a biometric signal for access control[1].

Speaker verification has been widely studied in domains such as telephony, smart assistants, and call centers. However, its application in VR environments remains relatively underexplored, particularly in scenarios involving natural, conversational interaction[4]. Existing approaches typically rely on explicit enrollment phases, predefined passphrases, or visible authentication interfaces, which may conflict with the principles of immersion and presence that VR systems seek to preserve.

This demo introduces a conversational speaker verification system embedded in a VR access-control scenario, where authentication and enrollment emerge naturally from interaction with a virtual non-player character (NPC). Instead of requiring users to explicitly register or authenticate themselves, the NPC initiates a short greeting when the user approaches a virtual entrance. The user's spoken response is processed in real time to verify identity, enabling immediate access for recognized users. When the speaker is not recognized, the system seamlessly transitions into a conversational enrollment process, collecting speech samples without interrupting the immersive experience.

By integrating real-time speaker verification into conversational interaction, the demo embeds authentication directly into the narrative flow of VR. This enables secure yet natural access control for applications such as virtual workspaces, social environments, and restricted virtual areas.

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2 SYSTEM OVERVIEW

The system consists of a VR environment containing a virtual entrance guarded by an NPC. When a user approaches the gate, the NPC initiates a fixed greeting prompt. The user's response is captured via the VR headset microphone and processed by a real-time audio pipeline.

The system operates in two modes:

- **Verification mode:** If the speaker embedding extracted from the response matches an existing identity above a confidence threshold, the NPC recognizes the user and grants access.
- **Conversational enrollment mode:** If no match is found, the NPC initiates a short conversational exchange to collect additional speech samples and registers a new speaker identity.

Importantly, the same initial utterance is always used for verification, ensuring consistent conditions while remaining natural and unobtrusive.

3 SPEAKER VERIFICATION PIPELINE

Captured audio is continuously acquired from the VR headset microphone during user interaction with the NPC. To ensure robustness and low latency, the audio stream is first processed by a lightweight voice activity detection (VAD) module, which filters out silence and non-speech segments.

Speaker embeddings are extracted using a pre-trained ReDimNet-based[5] speaker verification model for text-independent representation learning. ReDimNet generates fixed-dimensional embeddings from short speech segments and has demonstrated superior robustness to domain shifts, including microphone, and environments variations[2], which is particularly important in VR environments characterized by heterogeneous devices and noisy acoustic conditions.

Speaker verification is performed by computing the cosine similarity between the embedding of the current utterance and stored speaker templates. If the similarity score exceeds a predefined threshold, the user is authenticated; otherwise, the system seamlessly transitions to a conversational enrollment mode.

During enrollment, embeddings from multiple user responses are aggregated to form a speaker profile, enabling implicit registration without explicit commands or predefined passphrases.

All processing stages operate with low latency, ensuring immediate NPC feedback and preserving real-time interaction within the virtual environment. The pipeline runs locally with sub-second response time, preserving interaction flow.

4 SYSTEM IMPLEMENTATION AND TOOLS

The proposed system integrates a set of lightweight components to support real-time conversational interaction and speaker-based authentication in a VR environment, with emphasis on low latency and seamless integration.

Automatic Speech Recognition (ASR): User speech is transcribed in real time using the Whisper ASR model[3]. The transcriptions are used to support dialogue control and to extract identity-related cues during conversational enrollment. Simple text-based matching techniques are applied to compare detected names against a predefined database of valid user names.

Speaker Data Management: Speaker embeddings and associated identities are stored in a SQL-based database, enabling efficient retrieval and persistence of speaker profiles across sessions.

Text-to-Speech (TTS): NPC responses are generated at runtime using a TTS engine, providing immediate and context-aware feedback following speaker verification or enrollment decisions.

VR Environment: The immersive environment and interaction logic are implemented in Unreal Engine 5, which handles audio

capture, NPC behavior, and the coordination between verification and enrollment modes.

Together, these components form a unified pipeline for real-time speaker verification and conversational interaction in VR.

Furthermore, after the demo is completed no audio is retained, only the user features and audio transcriptions, which will also be deleted a day later.

5 DEMO SCENARIO

During the live demo, attendees can:

1. Approach the virtual gate and interact with the NPC.
2. Experience immediate access when recognized.
3. Go through conversational enrollment when unrecognized.
4. Re-enter the environment to observe successful speaker recognition.

The demo highlights how biometric authentication can be integrated into VR without explicit commands, passwords, or visible security mechanisms.

6 CONCLUSION

We demonstrate that conversational speaker verification can provide seamless and intuitive access control in immersive environments. By embedding biometric authentication into natural dialogue with NPCs, the system preserves immersion while enabling secure identity recognition. This approach opens new possibilities for personalized and secure interactions in VR workspaces and social environments.

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